After the Decision: Learning to Adapt from Delayed Outcomes

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Abstract

Scientific agents often choose among several tools that return plausible but conflicting answers. The challenge is not only selecting a provider before its output is visible: correctness feedback is produced after termination, may be delayed, and must update later choices without inventing labels for unselected tools. We present MIRA-Relay, an outcome-aware routing loop that combines a frozen discriminative provider surface with a lightweight provider-local residual adapter and an explicit release ledger. At cold start the adapter is exactly the frozen selector; after a selected result is released, it updates only that provider's residual reliability and can move the next action. We evaluate a one-call BBB routing contract with public truth, four checkpoint-bound fresh providers, scaffold-disjoint confirmation molecules, selected-only feedback, and delays $D \in \{0, 1, 4\}$. On the natural fresh holdout, MIRA-Relay matches Frozen at 89.81% selected correctness; under a controlled evaluator-side label flip specified before confirmation and applied when the fit-best provider is selected, it reaches 87.56% versus 73.38% for Frozen, a +14.18-point paired gain with a 95% scaffold bootstrap interval of [+9.61,+19.03] points. The result isolates the claim: delayed selected-only feedback supports recovery from the declared drift, while a stationary fresh menu does not justify a generic online gain.

CCS Concepts

• **Computing methodologies** → **Multi-agent planning**; *Reinforcement learning*; Machine learning.

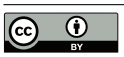
Keywords

tool selection, online learning, delayed feedback, scientific agents, molecular property prediction

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1 Introduction

A scientific agent must decide which tool's prediction to use when several providers estimate the same property. That choice matters even when every call completes: execution success establishes that an answer was returned, while usefulness depends on whether the answer is correct for the input. Molecular property prediction makes this distinction measurable because provider outputs can be checked against independently recorded endpoint labels.


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Valid responses can disagree on the same molecule. In our frozen BBB development panel, molecule `z000282` has benchmark label 1: `admet_ai_1_4_0` returns 0, while `b3clf_dtree_common` returns 1, both with successful runtime status. The example establishes a selection problem; it does not establish which provider is best on the next molecule. During evaluation, the agent sees neither unselected outputs nor the current correctness label before choosing.

Outcome feedback creates two obstacles: sparse coverage and delayed release. A fixed policy cannot use new evidence. A selected-only learner sees feedback only from chosen providers, leaving alternatives sparsely observed; delay postpones when feedback is usable. These constraints motivate our adaptation hypothesis but do not prove that any bandit fails. We therefore compare matched online controls and test whether released provider-local evidence changes later decisions.

We hypothesize that a released outcome can inform later choices through a shared released state. Each provider keeps its own residual, so the update need not assume that providers are independent or interchangeable. An immutable fit-time reference anchors the comparison. The claim requires a change in later exploitation under identical observations.

MIRA-Relay is the concrete method studied here. Its cold-start surface is frozen, while a selected-only reliability ledger adds a provider-specific residual only after the corresponding outcome is released. The residual is centered on that provider's fit-time success rate, so the adapter starts exactly at the frozen policy and changes only when released evidence is strong enough to move a later ranking. Runtime failure, unknown truth, and chemical error remain separate ledger fields; post-call verification and refusal are outside the primary one-call result.

Our contributions are threefold:

- We formulate scientific provider selection with independently recorded endpoint labels and delayed, selected-only feedback, separating correct answers from successful execution.
- We develop MIRA-Relay, which starts exactly at a frozen contextual selector and updates provider-specific residuals using two counters per provider. Its score-margin condition makes the effect of released evidence on later choices explicit.
- We evaluate the method against fixed, contextual-bandit, posterior, and gradient-update controls. On the stationary fresh panel it matches Frozen; under controlled correctness drift it improves success by 14.18 percentage points. Paired error counts and development interventions distinguish this recovery from general online improvement.

The experiments test both whether feedback changes a later choice and whether that change improves correctness. Development diagnostics then examine the score scale, residual centering, and offline objective that determine the starting point for adaptation.

2 Related Work

Tool-using agents learn to call external APIs and execute actions, but their training objectives generally do not impose a selected-only

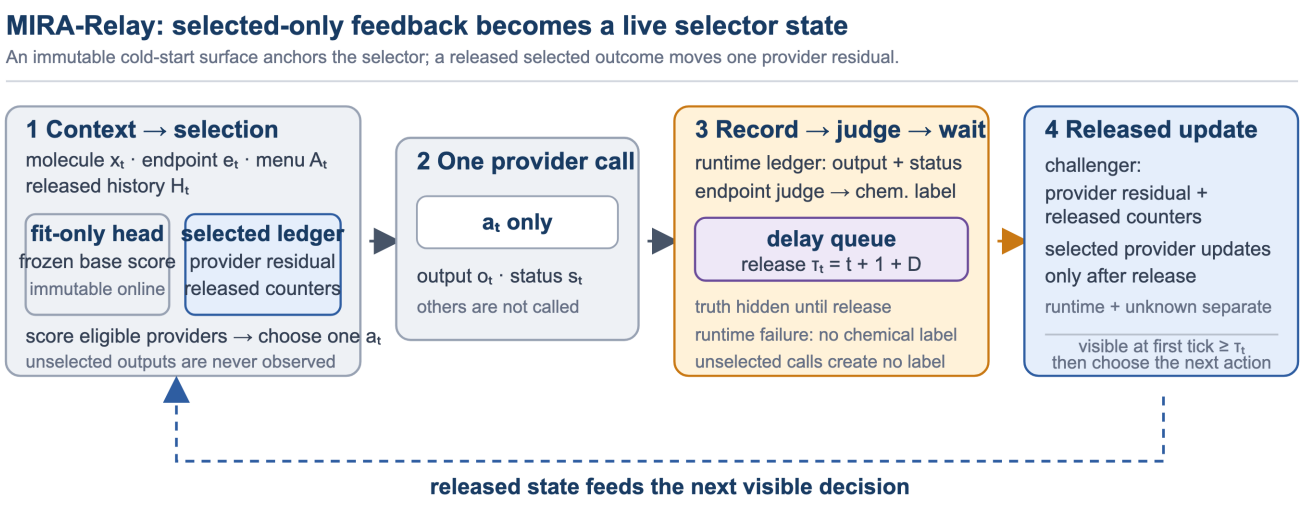


Figure 1: MIRA-Relay closes the loop from a pre-call choice to a later decision. Molecule, endpoint, eligible menu, and released state enter a frozen provider surface plus a provider-local residual ledger. One selected call produces a terminal record; the frozen endpoint judge supplies chemical correctness only after termination, while runtime failure follows its own ledger. The selected outcome waits until $\tau_t = t + 1 + D$ and then updates the released state; decisions continue while feedback is pending.

correctness clock [11, 14]. Our task instead treats the provider call as an online decision whose outcome may be delayed and whose correctness is tied to an independently auditable scientific endpoint.

Contextual bandits provide the natural language for choosing an action from a context before observing its reward. LinUCB [7] and contextual Thompson sampling [1] are canonical controls, while warm-start methods study how prior data can reduce the cost of early exploration [8]. Delayed-feedback analyses show that the release clock changes online learning even when the action space is unchanged [6]. MIRA-Relay adapts these ideas to scientific correctness by binding a provider-local residual to selected-only released outcomes and an explicit release clock; the same contract is applied to every comparator so that later actions have a common information boundary.

Recent model-routing work studies static or online selection among language models, including budget-aware adaptive routing [9], graph-based selection [3], dual-contrastive routing [2], and benchmark frameworks for multi-model routing [4]. Those systems provide useful modern comparators, but their usual feedback is a preference or task score for the selected response and their provider panels are language-model panels. We retain adapted versions only when their input features, prior, feedback, and budget can be audited under the pharmaceutical contract; GraphRouter is audit-only if its graph input semantics cannot be preserved.

Finally, molecular benchmark work emphasizes public task definitions and reproducible datasets [5, 13]. Pharmaceutical providers are correlated predictors with different training lineages; we therefore treat family sensitivity and training-membership as validity limits rather than treating every cached model as an independent truth source.

3 Problem Setting

We study one agent making a sequence of tool calls. Figures 1 and 2 show the same contract at two scales: the first separates the learned relay from execution and evaluation, while the second fixes the event order that determines when feedback can affect a later action. The contract is defined at the granularity of a concrete provider, not a provider family or a post-hoc majority vote.

3.1 Episode, menu, and feedback

At decision tick t, the agent receives a molecule x_t, an endpoint e_t, and an available menu A_t of concrete provider identifiers. It chooses one provider $a_t \in A_t$ before reading the current output panel. The benchmark stores an independent endpoint truth $y_t \in \{0, 1\}$ for evaluation. The selected call returns an output o_t, a runtime status $s_t \in \{\text{ok}, \text{error}\}$, and, when available, a confidence value c_t. We keep the labels that drive the method separate:

$$r_t^{\text{run}} = \mathbf{1}[s_t = \text{ok}],$$

$$r_t^{\text{chem}} = \begin{cases} \mathbf{1}[o_t = y_t], & s_t = \text{ok}, \\ \text{null}, & s_t = \text{error}, \end{cases}$$

$$r_t^{\text{sys}} = \mathbf{1}[s_t = \text{ok} \wedge o_t = y_t].$$

The studied correctness head uses only non-null r^{chem}. Runtime status is recorded separately; the reported fresh panel contains valid scored returns, so chemical and system correctness coincide there. We do not train a runtime prediction head in this experiment.

After termination, a replay judge records the selected outcome. The outcome becomes visible only at its registered release tick τ_t. The legal history at tick t is

$$H_t = \{(x_s, e_s, a_s, r_s^{\text{chem}}, r_s^{\text{run}}, \tau_s) : s < t,\ \tau_s \le t\}.$$

For example, for molecule `z000282`, the benchmark truth is 1. The call to `admet_ai_1_4_0` returns 0 with status 'ok'. The call to `b3clf_dtree_common` returns 1 with status 'ok'. Both calls have $r^{\text{run}} = 1$, but their chemical labels differ. A decision at tick 2 cannot read either unselected output, the current truth, or any future terminal record.

Provider parameters use a registry $\mathcal{P}$ fixed at initialization; $A_t \subseteq \mathcal{P}$ is the current candidate menu. Removing a challenger leaves its stored state intact. For the historical BBB replay, $|\mathcal{P}| = 33$ throughout the stream. The queue orders releases by due tick and breaks ties by enqueue order; each feedback is consumed once and receipts record the release order explicitly.

For the fixed-delay replay, a decision at tick t terminates, enqueues its feedback, and releases it at

$$\tau_t = t + 1 + D, \qquad D \in \{0, 1, 4\}.$$

Only after this release may the event enter $H_{t'}$ and affect a later decision. For $D = 0$, the release is processed before the next decision tick and never changes the action that caused the event. Unknown truth creates no correctness label and stops the current scoring stream; a refusal creates no selected-provider label. Both remain explicit in the episode ledger.

3.2 Released state and legal decisions

The online learner maintains a serializable post-release state S_t. At tick t, the release phase first applies exactly the events due at that tick; the selector then reads the resulting state:

$$\begin{aligned} S_t &= U\left(S_{t-1},\right. \\ &\qquad \left.\{(x_s, e_s, a_s, r_s^{\text{chem}}, r_s^{\text{run}}, \tau_s) : s < t,\ \tau_s = t\}\right), \\ a_t &= \pi_{\text{sel}}(x_t, e_t, A_t, H_t, S_t). \end{aligned}$$

With no due event, U is the identity. Thus a $D = 0$ event from tick t enters S_{t+1} and can affect action $t + 1$, but never action t. No correctness-residual update is allowed for an unselected provider, an unreleased event, an unknown label, or a refusal. A terminal unknown or refusal may still update runtime/availability bookkeeping; every selected scored provider, including the fit-best provider, receives only its own released residual update. This contract lets a receipt verifier check the visible read set, action, release queue, recorded state fields, and schema/counts. A separate receipt-closure audit materializes canonical state bytes from the sealed replay, independently recomputes scores and lifecycle metrics, and exercises checkpoint, logical-clock, and mutation checks. It audits the declared policy replay; it is not a new provider inference or effect estimate.

3.3 Estimands and costs

The primary estimand is prequential system success: a selected call must return a valid and correct answer. We also report conditional correctness, rescued and harmed decisions, failures, and update costs. The extension interface additionally accounts for escalation, refusal, and unknown truth; those utilities are outside the reported one-call result. Static full-panel accuracy and an evaluator-only retrospective best-fixed control measure task headroom separately from online effects.

4 MIRA-Relay

MIRA-Relay combines a frozen contextual score with a provider-specific correction from released outcomes.

4.1 Cold-start scores and the immutable reference

Let $\phi(x) \in \mathbb{R}^d$ be a frozen molecule representation and let $K = |\mathcal{P}|$ be the number of concrete providers. For each provider a, a logistic correctness head w_a^0 is fitted once on the fit partition. Its cold-start score is

$$q_a^0(x) = \sigma\big((w_a^0)^\top \phi(x)\big).$$

The current menu only masks which coordinates can act; no current output is read before the action. For the fresh BBB confirmation, $d = 65$: 64 Morgan bits plus an intercept. The fit-only Best Fixed provider is the provider with the highest fit-partition correctness. The frozen base scores are immutable throughout evaluation, and every provider switch is still counted even when it is wrong.

The running case is concrete. For molecule `z000282`, $\phi(x)$ is the frozen 65-dimensional vector built from its canonical SMILES and the intercept; the fit heads assign separate scores to the four fresh providers. At cold start, MIRA-Relay and Frozen therefore choose the same provider for the same menu. Providers outside the sealed checkpoint manifest are out of scope.

4.2 Context-relative reliability adapter

For provider a, let $\bar{r}_a$ be its fit-partition correctness rate and let $n_{a,t}$ and $s_{a,t}$ be, respectively, the number of its selected outcomes released by tick t and the number that were correct. With a fixed prior count $n_0 = 20$, the released posterior rate and residual are

$$\widehat{r}_{a,t} = \frac{n_0 \bar{r}_a + s_{a,t}}{n_0 + n_{a,t}}, \qquad \Delta_{a,t} = \widehat{r}_{a,t} - \bar{r}_a.$$

The action score is

$$q_{a,t}(x) = q_a^0(x) + \Delta_{a,t}, \qquad a_t = \arg\max_{a \in A_t} q_{a,t}(x_t).$$

At cold start, $n_{a,0} = s_{a,0} = 0$ and therefore $\Delta_{a,0} = 0$: the adapter is exactly Frozen. When a selected outcome becomes visible, only that provider's counters change; an unselected provider never receives a synthetic label. A runtime error is recorded as an execution failure and is not inserted into the chemical-correctness count. This provider-local residual is added to the x-conditioned frozen score. Here "context-relative" refers to the current provider ranking and released evidence state, not to a second x-conditioned head; the fit-time representation itself never changes.

The case makes the update concrete. If the selected provider for `z000282` is correct, $s_{a,t}$ and $n_{a,t}$ each increase by one at its release tick. If it is wrong, only $n_{a,t}$ increases. The next score changes by the corresponding residual; all unselected providers keep their previous residuals. The update is $O(1)$ per released event and scoring is $O(Kd)$, while the online ledger stores $2K$ counters and the fixed heads remain outside the online state.

4.3 Release ledger and delayed update

The online state is the selected-only ledger $S_t = \{(n_{a,t}, s_{a,t}) : a \in \mathcal{P}\}$. At tick t, the release queue first applies every event with due time $\tau = t$; the selector then reads the resulting state. Releases

follow the delay rule in Section 3: feedback from tick t is visible from $t + 1 + D$. The released event prefix determines the counter updates and the next score.

The update touches only the selected provider. For a scored terminal record s released at tick $t = \tau_s$, with chemical label $r_s^{\text{chem}} \in \{0, 1\}$,

$$n^+_{a_s,t} = n^-_{a_s,t} + 1, \qquad s^+_{a_s,t} = s^-_{a_s,t} + r_s^{\text{chem}},$$

where − and + denote the state immediately before and after the queue applies that release; all other provider counters are unchanged. If the call fails, r^{chem} is null and the runtime-failure ledger records the event separately. Unknown truth and refusal do not update correctness counters. This separation is the central information boundary: the learner receives exactly the selected outcome after release, never the current truth or an unselected provider's output.

The update is $O(1)$ per released event and scoring all K providers costs $O(Kd)$. The online state uses $2K$ counters plus masks and the release queue. For `z000282`, a released result changes one provider residual, so the next action can switch according to the evidence.

4.4 Action rule and falsification

The primary one-call runner uses the highest current score in the eligible menu. There is no result-dependent probe in the reported fresh confirmation, so every method receives the same one-call budget and the same selected-only feedback clock. The controlled stress flips the selected label of the fit-best provider after the specified midpoint; it does not reveal any unselected output or alter the input features.

The implementation preserves five invariants: (i) no current truth or unselected output is read before action, (ii) only released selected events update state, (iii) the fit surface is immutable, (iv) runtime failure and chemical correctness remain separate, and (v) the released event prefix determines the next action. The receipt records a learned-state snapshot, release queue, action, and event schema; an independent receipt checker validates the exact event identities and order, selected correctness, release clocks, and state-hash/version continuity. The v3 closure artifact adds canonical state bytes and terminal/final hashes for the declared replay and is checked independently.

The adaptive claim is rejected if the residual cannot change a later action under the declared drift, if contract/replay checks find a mismatch, or if the gain disappears against the same-schedule Frozen and matched Online-SGD controls. Action changes alone are not treated as utility; the evaluation reports rescued and harmed decisions, scaffold-cluster intervals, calls, and update latency.

5 Experiments

5.1 Two-layer benchmark contract

The benchmark has a public truth layer and a provider-routing layer. The truth layer uses the Therapeutics Data Commons (TDC) BBB endpoint, fixed label semantics, and its split archive [5]. The development manifest contains 722 labeled molecules; 716 have complete outputs from the 33-provider Task36 panel. Conflict groups are excluded under the project decision, with raw receipts retained for audit. Because the old panel overlaps the official BBB test (147 canonical molecules; 235 scaffolds), it is not an untouched confirmation set.

Table 1: Comparator groups under the common routing contract.

Role	Methods	Purpose
Anchors	Random; Best Fixed; Retrospective Best Fixed[†]	blind, fit-only and evaluator-only references
Frozen / matched updates	Frozen; MIRA-Relative; Online SGD	isolate the value of a released update
Exploratory diagnostic	pooled-feedback static recentering[‡]	report an implementation boundary, not a mechanism baseline
Mechanism control	fixed cyclic permutation credit[§]	isolate source-aligned credit under the declared drift
Modern online controls	Greedy LinUCB; BayesianRouter	current contextual and posterior controls
Proposed	MIRA-Relay	selected-only provider residual under delay

[†]Evaluator-only retrospective control; it is not a deployable baseline.
[‡]The pooled term is common across providers and cancels from the arg max; the diagnostic is excluded from mechanism claims.
[§]A sealed-receipt CPU replay with the same cold start and release clock; it is a mechanism control, not a deployable baseline.

The routing layer turns cached outputs into sequential decisions with a fixed registry, eligible-menu mask, selected-only feedback, delayed release, and explicit failures. It is a project protocol grounded in public truth. The fresh study keeps four checkpoint-bound providers; the wider 33-provider pool is development evidence, not a model-fresh claim. Unknown truth stops a single-call stream, while verification, refusal, provider arrival, hERG, and AMES use separate extension contracts.

5.2 Baselines and fairness

Table 1 groups the deployable comparators by the role they play in the routing question. Every method receives the same molecule, endpoint, menu, fit-only initialization, selected-only feedback, delay, and one-call budget. Full-panel labels are used only for fit warm starts or evaluator-only controls. Greedy LinUCB [10] is adapted to a single-step horizon; BayesianRouter [12] uses a task-specific prior and reward interface.

MIRA-Relative is a stopped negative control: it multiplies each provider's frozen score by a molecule-independent (x-independent) factor, rather than adding the provider-local residual used by MIRA-Relay. Its natural run therefore tests whether selected-only multiplicative factors can change later rankings. It is retained as an ablation, not as a second proposed method.

We retain a pooled-feedback diagnostic only as an implementation audit. Its pooled term is common across providers and cancels in the arg max, leaving a static fit-rate recentering; its initial ranking also differs from Frozen. It is not a matched online wrong-credit baseline or mechanism evidence. The raw receipt remains in the artifact.

Recent routers with incompatible input or feedback semantics remain audit-only. PILOT, GraphRouter, RouterDC, and RouterBench need a separate wrapper and leakage audit; our result covers the sealed four-provider menu, not 33-provider or model-fresh generalization.

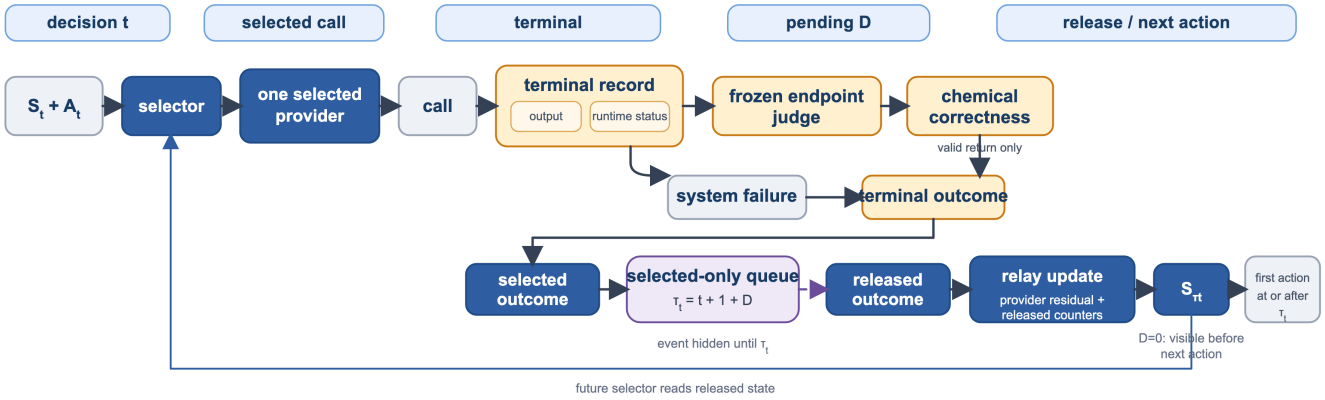


Figure 2: The terminal feedback contract. At decision tick t, the selector sees S_t and A_t and chooses one provider before any current output is observed. The terminal record is split into output and runtime status; the frozen endpoint judge then yields chemical correctness only after termination. Runtime failure and chemical correctness form the selected terminal outcome, which waits in a selected-only queue until $\tau_t = t + 1 + D$. The relay update is visible to the first action at or after release; for $D = 0$, this is the next action.

Table 2: Experimental map. E1–E2 test released feedback; E3–E4 diagnose the development surface; E5–E6 measure cost and post-call utility.

ID	Question	Evidence
E1	Adaptation and drift	Table 3, Figure 3
E2	Source-aligned credit	Table 4
E3	Initial selection	Table 8
E4	Scale and centering	Table 7
E5	Policy cost / validity	Table 5
E6	Post-call refusal	Table 6

5.3 Evaluation protocol

The fit partition is scaffold-disjoint from 157 confirmation molecules. Each policy sees only molecule features, endpoint, eligible menu, and released selected outcomes. Five folds, two seeds and $D \in \{0, 1, 4\}$ yield 30 schedules and 4,710 paired calls per method; each stream makes 157 calls. The primary natural matrix and controlled intervention were frozen before confirmation: after tick 78, the evaluator flips the selected fit-best provider's label, keeping features, menus and feedback visibility fixed. The primary result includes 270 natural and 150 stress streams. Paired intervals resample Bemis–Murcko scaffold clusters after aggregating repeated schedules.

The development diagnostics use 722 molecules and five scaffold folds. Their update factorial adds three delays (2,166 decisions per cell); offline objectives use paired training seeds. The post-call study reuses the confirmation cohort as a secondary analysis. Protocols and hyperparameters are in the artifact; Table 2 maps the six questions to their evidence.

5.4 Natural and controlled outcomes

Natural fresh menu. We ask whether released selected outcomes improve a stationary fresh menu. Table 3 shows that Frozen is within 0.64 points of the retrospective best-fixed reference. MIRA-Relay changes 1,656 actions, rescuing and harming 82 decisions each, so its 89.81% accuracy is identical to Frozen (paired $\Delta = 0.00$ points, 95% CI $[-1.24, +1.40]$). The BayesianRouter, Greedy LinUCB, and matched SGD controls likewise show no stationary gain. This is a stability check, not evidence for a generic online improvement.

Controlled drift recovery. Under the specified evaluator-side stress, Table 3 shows MIRA-Relay at 87.56% versus 73.38% for Frozen: +14.18 points (95% CI [+9.61, +19.03]). It changes 1,884 actions, rescuing 780 and harming 112 decisions relative to the same-receipt Frozen action. Online SGD, Greedy LinUCB and BayesianRouter recover by +10.66, +7.77 and +6.90 points. Figure 3 shows the paired gains at each delay. This response to a declared label flip is not evidence of natural provider nonstationarity. The pooled-feedback diagnostic is excluded because its common term does not give a matched wrong-credit intervention.

Mechanism attribution. The sealed-receipt control shares Frozen's cold-start scores and actions, then changes only the destination of released source residuals. Table 4 reports its separate equal-weight scaffold-mean estimand: 90 streams per condition, two deterministic schedule replicas and $D \in \{0, 1, 4\}$. Local–permutation is +15.67 pp under the evaluator-side drift; its natural interval crosses zero. This supports source-aligned selected-provider credit for the declared replay, without identifying stationary gain, family sharing, verification/refusal or agent gain.

Table 3: Pre-call selection on BBB: natural outcomes and recovery under controlled correctness drift.

	Natural	Controlled drift		
Method	Success (%)	Success (%)	Gain (pp)	95% CI
Random	83.76	—	—	—
Best Fixed	88.15	—	—	—
Frozen	**89.81**	73.38	reference	—
MIRA-Relative	**89.81**	—	—	—
Online SGD	89.21	84.03	+10.66	[+6.05, +15.45]
Greedy LinUCB	89.72	81.15	+7.77	[+5.08, +10.72]
BayesianRouter	89.53	80.28	+6.90	[+3.50, +10.41]
MIRA-Relay	**89.81**	**87.56**	+14.18	[+9.61, +19.03]
Retrospective Best Fixed†	90.45	—	—	—

Gain is paired against Frozen. † Evaluator-only fixed provider. Bold marks the best deployable point estimate, including ties. —: not in the recorded roster.

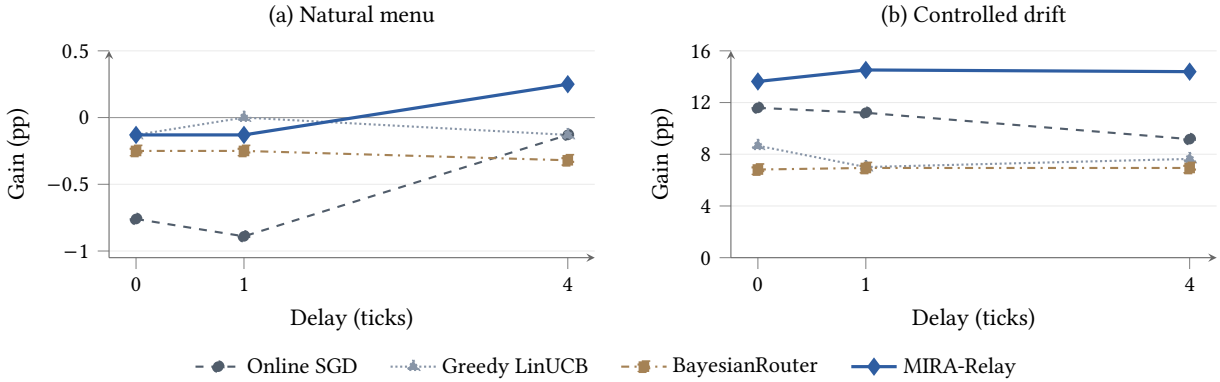


Figure 3: Delay sensitivity. Paired gains over Frozen at the recorded delays; points are descriptive, with no uncertainty shown.

Table 4: Source-aligned attribution with matched cold start and release clock.

Contrast	Natural Δ [95% CI]	Controlled Δ [95% CI]
Local − Frozen	+0.10 [−1.08, +1.42]	+14.40 [+9.79, +19.23]
Permutation − Frozen	+0.41 [−0.47, +1.32]	−1.27 [−3.66, +0.95]
Local − Permutation	−0.30 [−1.42, +0.76]	+15.67 [+10.65, +20.74]

Units: pp. Equal-weight scaffold-mean contrasts over 124 clusters; 90 replay streams per condition. This estimand differs from the pooled-call point estimates in Table 3.

5.5 Decision changes and costs

Rescued and harmed decisions distinguish valuable switches from damaging substitutions; execution failures, unknown truth and refusal retain separate statuses. Table 5 reports CPU policy overhead as the mean of 30 stream-level p95 timings per method and condition. The timer excludes file I/O, provider inference, evaluator scoring and service latency. Every stream uses one call per molecule; all 270/270 natural and 150/150 stress receipts pass contract/replay checks. These measurements do not establish end-to-end service latency or live learner memory.

5.6 Post-call risk and utility

We analyze post-call refusal on the same 157 held-out molecules used above. This is a secondary analysis: the cohort had been inspected before this experiment was designed. In each of the five development folds, the provider with highest fit accuracy makes one call. The gate accepts its answer when $2|p - 0.5|$ is at least a fit-only threshold, set to the $\lfloor 0.8 n_{\text{fit}} \rfloor$th largest fit confidence (including ties). Each comparator receives the same returned prediction.

Table 6 shows a 5.88-point accepted-accuracy gain over matched random (95% scaffold bootstrap CI [+2.77, +9.29]). At the fixed

Table 5: CPU policy overhead (μs; lower is better).

	Natural menu		Controlled
Method	Selection	Update	Update
Random	67.14	5.44	—
Best Fixed	5.82	0.68	—
Frozen	15.89	1.31	1.50
MIRA-Relative	42.02	73.30	—
Online SGD	14.08	19.37	20.02
Greedy LinUCB	128.63	90.90	92.32
BayesianRouter	133.65	88.96	92.55
MIRA-Relay	27.21	2.81	3.46

Mean of 30 stream-level p95 timings; macOS arm64. Excludes provider inference and I/O. All methods use one call per input. Live learner memory remains unmeasured.

utility setting below, the gate improves over always-accept by 0.0338 (95% CI [+0.0088, +0.0609]).

Same-coverage random is an evaluator-side conditional null: within each fold, it accepts exactly as many answers as the confidence gate, at randomly selected positions. Its realized coverage requires the complete cohort, so it is not an online baseline. All policies make one call before accepting or refusing; the five fitted gates reuse the same molecules. Bootstrap intervals resample 124 scaffold clusters with all repeated observations kept together and condition on the fitted gates and the fixed random draws.

We measure the trade-off with $U = -(W + \rho R + \kappa C)/N$, where W counts accepted wrong answers, R refusals, and C calls. A wrong answer costs one unit. The preregistered illustrative setting is $(\rho, \kappa) = (0.25, 0.01)$; it is not a clinical cost estimate. Since call counts are equal, κ cancels from every paired comparison. Relative to always-accept, the confidence gate removes 57 wrong answers at the cost of 122 refusals:

$$U_{\text{gate}} - U_{\text{accept}} = \frac{57 - 122\rho}{785}. \tag{1}$$

The observed advantage changes sign at $\rho = 57/122 \approx 0.467$. Against refusal after every call, its break-even cost is $36/663 \approx 0.054$. These are point-estimate utility boundaries, not risk guarantees. Testing a learned accept/verify/refuse decision within the delayed-feedback routing loop requires an integrated experiment.

5.7 What does the residual correct?

The natural run is a stationary stability check: MIRA-Relay changes 1,656 actions, but 82 rescued and 82 harmed decisions cancel in aggregate. Under the declared evaluator-side drift, released selected outcomes change later actions and improve correctness relative to Frozen; the matched cyclic replay supports source-aligned provider credit as the reason for this declared effect. The four fresh providers are singleton families, so family-shared versus independent residuals remain unidentified, and the result does not establish a universal drift response.

Development diagnostic. Summing only over released selections of provider a, the residual decomposes exactly as

$$\Delta_{a,t} = \frac{\sum(r_s - q_a^0(x_s))}{n_0 + n_{a,t}} + \frac{\sum(q_a^0(x_s) - \bar{r}_a)}{n_0 + n_{a,t}}. \tag{2}$$

Independent reconstruction of 2,166 development decisions across five folds and three delays agrees with recorded scores within 2.23×10^{-16}. Omitting the second term on the unchanged selected histories changes 401 shadow actions (18.51%). This measures sensitivity, not the accuracy of an alternative policy. Class-balanced training also means that frozen sigmoid scores need not be calibrated correctness probabilities; the two terms cannot be identified solely with true drift and selection bias.

We therefore ran a development factorial separating the score scale from the update rule (Table 7). Inverse class-weight correction adds $\text{logit}(\bar{r}_a)$ to each frozen logit; query centering replaces $\bar{r}_a$ in the released residual by the score saved at that decision. Each variant follows its own selected-feedback history and has a matched Frozen control. The correction lowers OOF Brier error from 0.235 to 0.173, but corrected query centering adds only 0.14 percentage points over its Frozen control on the natural stream. Its gains over corrected marginal centering are 0.51 and 0.37 points, below the predeclared one-point development threshold. Better score scaling alone therefore does not establish a substantial online benefit.

Corrected Frozen selects the stress target on 56.91% of post-change calls, versus 34.81% for raw Frozen. Cross-score recovery increments thus also reflect different exposure to the intervention; they cannot isolate the update mechanism. The original method and confirmation results remain fixed throughout this diagnosis.

Where can better initial selection help? The four-provider development panel separates shared errors from selection errors. Of 722 molecules, all four providers are correct on 471, disagree in correctness on 147, and are all wrong on 104. A per-molecule evaluator oracle therefore reaches 618/722 (85.60%), while corrected Frozen reaches 579/722 (80.19%). The remaining 39 recoverable errors define selection headroom under this menu; the 104 shared errors cannot be repaired by choosing another member of it. This oracle uses unselected outputs solely for diagnosis and differs from the retrospective best-fixed reference in the confirmation table. These counts motivate improving the conditional ranking without claiming that every available alternative is predictable before a call.

Does changing the offline objective improve initial selection? We test two offline objectives on this four-provider development panel (Table 8). Both use normalized Morgan64 inputs, a 64-unit ReLU encoder, and identical score and auxiliary heads. The control minimizes score BCE plus 0.2 auxiliary BCE; the alternative uses the BayesianRouter-style correct–wrong Bradley–Terry objective plus the same auxiliary loss. Each of 20 fits runs 200 full-batch AdamW steps from paired initial weights. Checkpoints maximize validation selection accuracy, with earliest ties. Neither objective improves on corrected Frozen in either seed. All five leave-one-fold-out differences are negative for each comparison. Under this representation and training recipe, replacing the score objective does not recover the remaining routing headroom.

Table 6: Post-call refusal on the reused BBB holdout (secondary analysis).

Policy	Coverage (%)	Accepted accuracy (%)	Utility
Always accept	100.00	88.15	−0.1285
Confidence refuse	84.46	**94.57**	**−0.0947**
Always refuse	0.00	—	−0.2600
Same-coverage random†	84.46	88.69	−0.1444

† Evaluator-side conditional null. $U = -(W + 0.25R + 0.01C)/N$: W accepted errors, R refusals, C calls, N records. All policies call once; bold marks the best deployable point estimate.

Table 7: Score scale and update rule on BBB development. Accuracy in %.

Score	Update	Natural	Drift
Raw	Frozen	79.36	69.11
	Marginal	79.18	77.29
	Query	78.49	75.95
Corrected	Frozen	80.19	63.30
	Marginal	79.82	78.30
	Query	80.33	78.67

Compare update rules within each scale; score correction also changes exposure to the intervention.

Table 8: Cold-start accuracy on BBB development. The per-molecule oracle is evaluator-only.

Selector	Accuracy (%)
Best Fixed	79.22
Raw Frozen	79.36
Corrected Frozen	**80.19**
BCE + auxiliary CLS	78.53 / 78.95
BT + auxiliary CLS	78.81 / 78.25
Per-molecule oracle†	85.60

Paired training seeds 0/1 are separated by '/'; one value denotes a deterministic result. † Uses unselected outputs.

6 Discussion and Limitations

The benchmark separates public answer semantics from the project routing protocol. This makes the provider-selection question replayable, but it does not make the routing layer a community benchmark by itself. The principal development endpoint is BBB; hERG and AMES remain extensions until their assay semantics, coverage, conflicts, and provider training lineage are closed. The current provider panel also contains related model families, so provider-count improvements cannot be interpreted as independent evidence without family-balanced analysis.

Results are scoped to feedback-adaptive selection: delayed replay does not establish natural provider arrival, retirement, version drift, or broad nonstationarity. The controlled stress is not deployment evidence. Frozen is a reference and guardrail, not a no-harm guarantee.

Failure can arise from sparse feedback, sub-margin updates, or error transfer. We therefore report exposure, score margins, and rescued/harmed switches: action changes show response, while rescued and harmed decisions show its value.

6.1 Validity and reproducibility

The confirmation is separated from the development diagnostic at three points. First, provider checkpoints are sealed before the candidate molecules are scored; the four fresh models are bound to their source commit, checkpoint hash, and native prediction code. The 157 confirmation molecules are scaffold-disjoint from the development fit and validation partitions. The truth label is used only by the evaluator after a selected provider returns, never by the selector at decision time.

Second, every stream is recorded as a receipt rather than as an aggregate score. A decision record contains the molecule identifier, menu, feature hash, released-state version, chosen provider, and action score; it contains neither the current truth nor an unselected output. A terminal record binds the chosen provider to its returned output and runtime status. The release queue records the due tick and consumption order. The serialized learned state, including provider counters and prior count, is hashed after the corresponding selected feedback is released. An independent checker validates the receipt contract and detects event, clock, binding, or commitment-chain inconsistencies; the v3 closure artifact materializes the canonical state store and replays it independently, including final-state and terminal commitments.

Third, uncertainty is clustered by molecule scaffold. The natural result pools 5 folds, 2 seeds, and 3 delays over the same 157 confirmation molecules, while the controlled stress uses the same schedule and changes only the selected label of the fit-best provider after tick 78. The reported intervals are bootstrap intervals over Bemis–Murcko scaffold clusters; they are not intervals over individual event rows. We report both rescued and harmed switches because a positive aggregate can hide unsafe substitutions. Contract/replay checks found 270/270 natural streams and 150/150 stress streams replayable with no forbidden decision fields or action mismatches.

This design also fixes what the result cannot establish. The natural fresh menu is a stationary checkpoint panel, so its null online gain is evidence against a generic online improvement claim, not evidence against adaptation under every kind of drift. The stress

is a mechanism test with a declared evaluator-side label flip, not a deployment trace. The post-call verification result uses a separate accept/refuse utility contract and does not establish provider arrival, escalation, or hERG/AMES generalization.

The event ledger makes premature feedback use detectable: every decision must bind to exactly one terminal outcome and one legal release. Methods share the molecule order and delay schedule. For the six-cell development diagnostic, complete counter and pending-score states are also recorded; an independent replayer reconstructs all 77,976 events, their choices, and paired aggregates. The v3 closure audit supplies canonical state bytes for this declared replay; it does not turn the sealed provider predictions or offline fit into a second training or inference implementation.

6.2 Artifact scope

The artifact is organized around the routing contract. Truth and provider manifests bind molecule identifiers, endpoint labels, splits, checkpoints, commits, feature code, and hashes; terminal records keep evaluator truth and provider outputs after the action.

The static diagnostic includes 30 initialization/best-checkpoint files, 4,000 epoch records, and 2,888 held-out predictions. Independent NumPy recovery matches both heads, choices, aggregates, and checkpoint selection; maximum GPU-to-NumPy logit error is 1.93×10^{-7}.

Runtime error, unknown truth, and refusal retain separate status and utility semantics. The primary one-call streams contain no such events. The secondary refusal analysis records accept/refuse actions and evaluator outcomes separately; multi-call and menu-change policies require further transitions.

An agent can consume the selected output and its released reliability record. The routing and refusal experiments assess separate decisions; an integrated agent must additionally be tested on final answers and paid tool calls.

7 Conclusion

Adapting scientific tool use requires connecting a released outcome to a better subsequent decision. MIRA-Relay makes that connection testable through a frozen provider surface, a selected-only provider residual, and an explicit release clock. On a natural fresh menu it matches the frozen selector; under the declared correctness drift it recovers more accurately than the matched controls. The remaining evidence-extension and lifecycle claims are intentionally outside this paper.

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